Kush Agrawal

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EDUCATION

International Institute of Information Technology, Naya Raipur Aug 2024-Aug 2028
B.Tech in Data Science and Artificial Intelligence **CGPA: 8.22/10**

TECHNICAL SKILLS

Programming: Python, Bash, TypeScript, SQL, C++; data structures and algorithms.
Systems and Networking: Linux/Debian, systemd, Proxmox VE, TCP/IP, DNS, HTTP/HTTPS, SSH, Nginx, Cloudflare Tunnel.
Cloud and Platform: Docker, Docker Compose, Kubernetes, AWS, GitHub Actions, Terraform, Ansible.
Observability and Reliability: Prometheus, Grafana, Monit, collectd; monitoring and alerting, SLIs/SLOs, error budgets, incident response, root cause analysis.
Data and Storage: PostgreSQL, MySQL, SMB/CIFS, rsync, Btrfs RAID1.

INFRASTRUCTURE AND SYSTEMS PROJECTS

Homelab/Production Server probablysecure.tech

Debian, Docker Compose, Nginx, OAuth2 Proxy, Cloudflare Tunnel, Python, systemd

- Operate a self-hosted production environment for remote media streaming and file access, running **nine containerized services** across a NAS and Raspberry Pi behind authenticated web access.
- Automated full-stack recovery after host power loss using systemd, Wake-on-LAN, and campus-session renewal, reducing service recovery from **about 20 minutes of manual intervention to 150 seconds**.
- Added proactive monitoring and alerting for **service availability, disk utilization, and TLS certificate expiry**; regularly test power-loss recovery and backup restoration.
- Diagnosed duplicate-IP outages using ARP responses and moved campus addressing to DHCP; ran an x86-only authentication client on ARM64 using **QEMU** with scheduled connectivity checks.

SENTINEL: OSINT Platform GitHub

Python, Flask, MySQL, Docker Compose, Nginx, GitHub Actions, AWS ECR/EC2/SSM

- Built an investigation platform for subscriber-record lookups, bulk imports, and audited reports; made large imports resumable with **database checkpoints, streaming reads, batched inserts, and dry-run validation**.
- Automated container delivery through GitHub Actions, OIDC, ECR, and SSM with **deployment locks, health checks, and rollback paths** to reduce deployment failure risk.

Adaptive SSH Honeypot and Threat Analysis

Python, Paramiko, Kubernetes, Proxmox VE, Grafana, Implicit Q-Learning, Gemini API

- Built an SSH decoy that simulates a Linux terminal and captures authentication attempts, commands, source IPs, and timestamps for attacker-behavior analysis.
- Deployed honeypot components using **Kubernetes on a Proxmox cluster** and built **Grafana dashboards** for service and attack telemetry; added bounded ML/LLM retries, validation, and fallback responses.

EXPERIENCE

Université Polytechnique Hauts-de-France Jun 2026-Present
Machine Learning Research Intern | Campus visit: France

- Built a Python/Pandas pipeline integrating **7 EV-charging datasets into 261K+ hourly observations** and developed a two-stage XGBoost forecasting model using chronological holdout evaluation and time-series cross-validation.

Indian Institute of Technology, Patna May 2026-Jul 2026
Cybersecurity Research Intern

- Implemented the PQC-DLST-MQTT prototype using ML-KEM, ML-DSA, AES-GCM, and SHAKE-256 with deterministic multi-publisher nonces and out-of-band key management.

Indian Institute of Information Technology, Nagpur Nov 2025-Feb 2026
Network Research Intern

- Built Gymnasium environments and PPO policies for network resource allocation, modeling finite-buffer packet loss and energy/carbon trade-offs using Python and Stable-Baselines3.

PATENT

Quantum-Secured IoT Air-Pollution Monitoring Indian Patent No. 589285

Co-inventor | Granted: 13 May 2026 Patent certificate
Designed hybrid blockchain/IPFS storage and integrated QKD with AES encryption for secure IoT data handling.